

Christof Teuscher

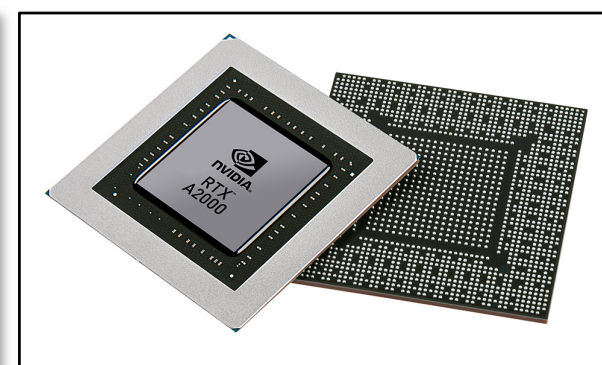
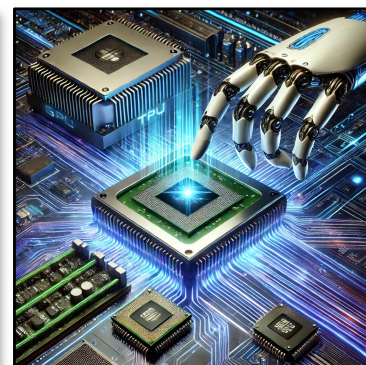
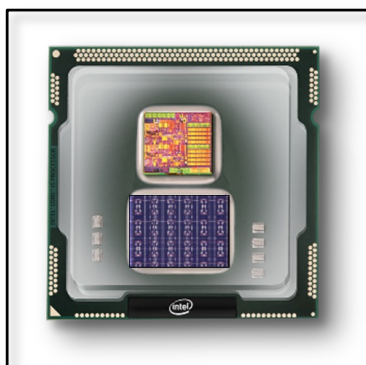
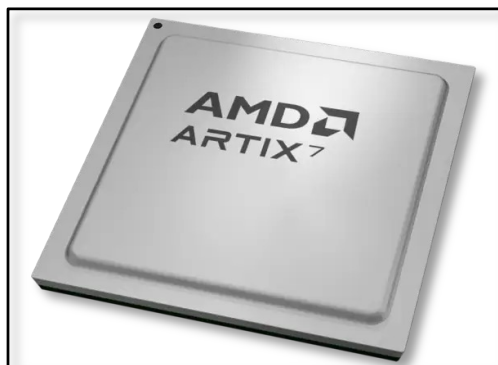
ECE 410/510: Hardware for AI and ML, Spring 2026

Week 5: Recap, outlook, and reminders

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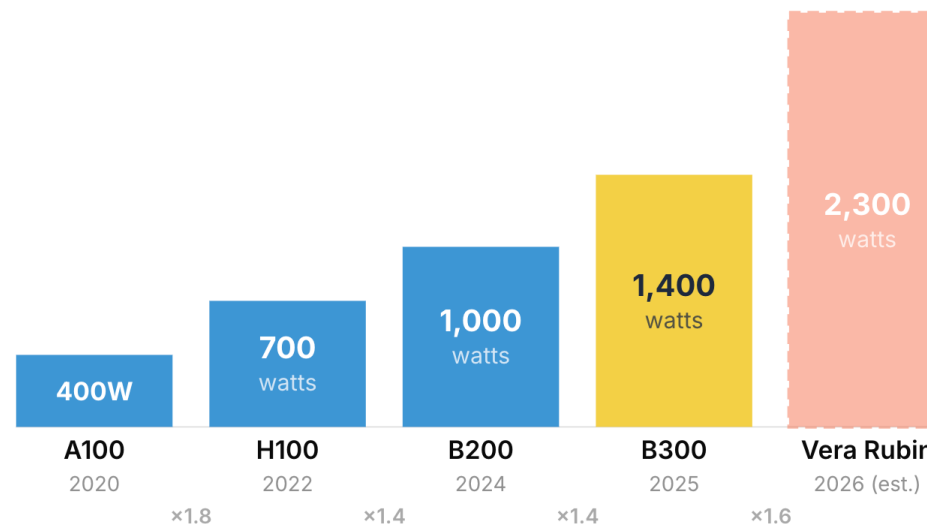


What did you learn last week?

What a single GPU draws

GPU power is measured in TDP (thermal design power), the maximum sustained wattage the chip draws when working as hard as it can. Training runs hold GPUs at 70-90% of their TDP for weeks or months at a time. Most traditional servers average only 20-50% of their rated power.^[2]

NVIDIA releases a new generation of data center GPUs roughly every two years, and each generation draws significantly more power. The A100 (2020) drew 400W (watts). The H100 (2022) draws 700W. The B200 (2024) draws 1,000W in its air-cooled configuration and up to 1,200W when liquid-cooled.^{[3][4]} The B300 draws 1,400W.^[5] NVIDIA's next architecture, Vera Rubin, targets 2,300W per GPU.^{[6][7]}



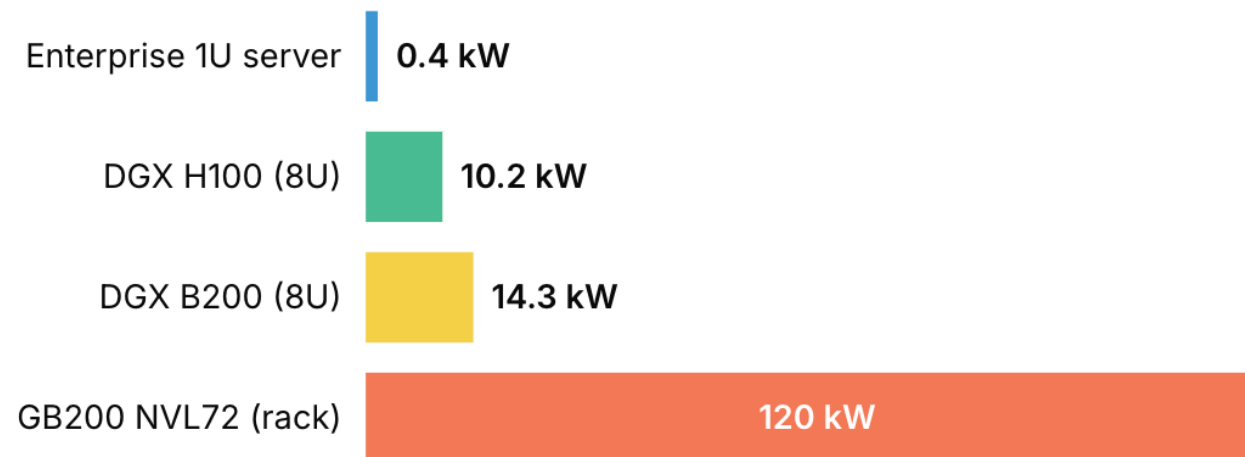
NVIDIA training GPU TDP across generations [3][5][6][7]. Vera Rubin is a roadmap estimate.

A regular house typically requires between **5,000 and 10,000 watts** to power essential appliances simultaneously.

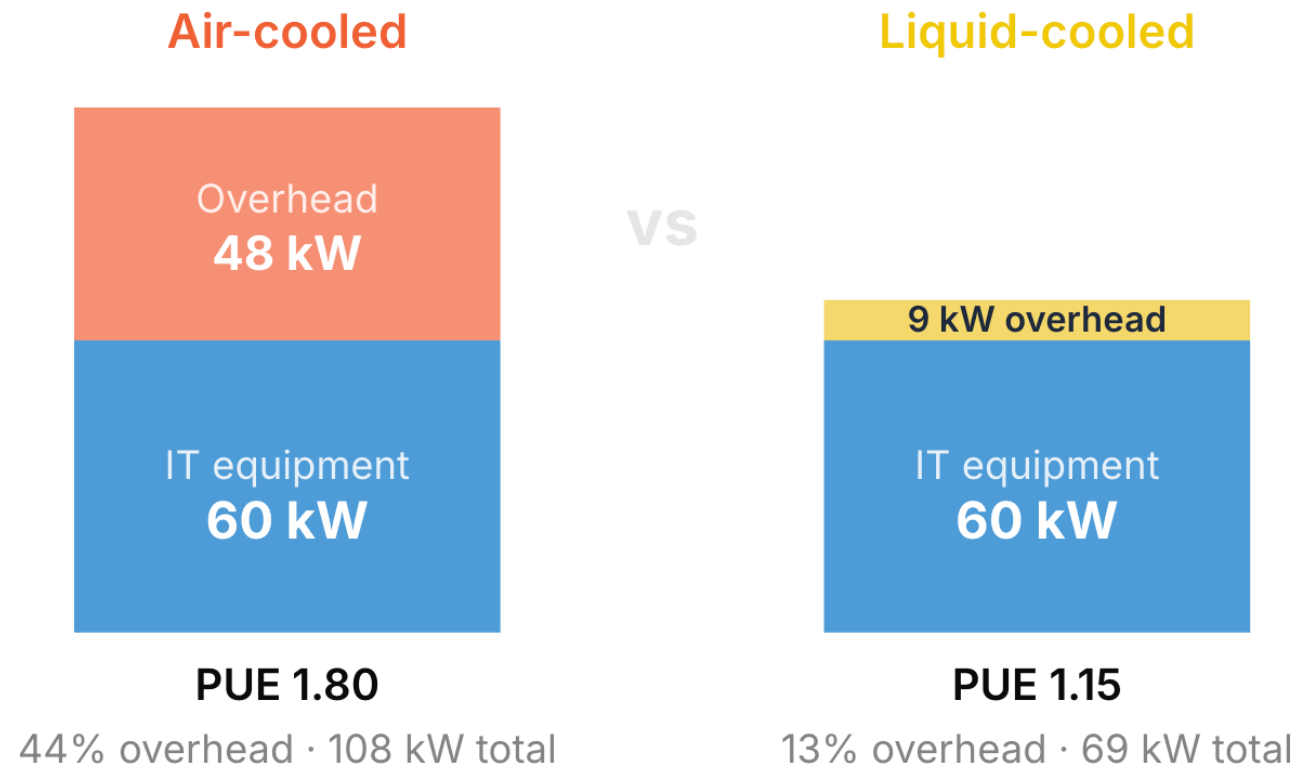


From server to cluster

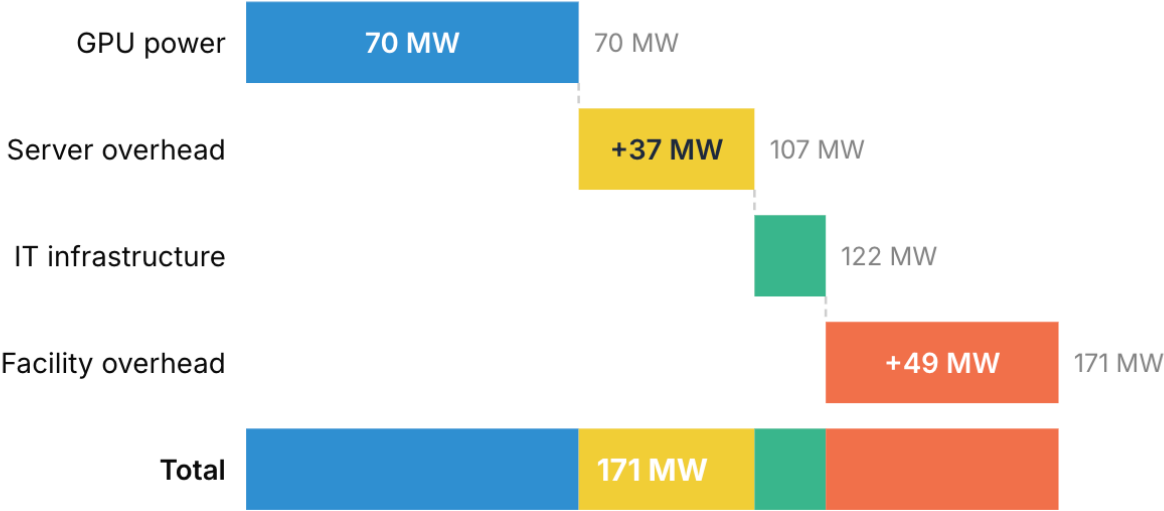
A typical single-unit server running web or database workloads draws 300-500 watts.
The AI systems in the previous section draw 20x to over 200x that.



Power per system, from a single enterprise server to the GB200 NVL72 rack-scale system [14]. The NVL72 is an integrated rack, not a collection of discrete servers.



Both bars show the same 60 kW IT load. PUE values from Uptime Institute [21].



Illustrative power multiplier stack for 70 MW of GPU power. Multipliers from Epoch AI [40], derived from GB200 NVL72 reference architecture.

Facility	Power	Status
xAI Colossus, Memphis	150-250 MW (100K→200K GPUs) [16] [17] [18]	Operating
Stargate (OpenAI/SoftBank/Oracle)	Up to 10 GW across multiple sites [41]	Announced January 2025
Meta total fleet (2023)	~1.7 GW continuous (14,975 GWh/yr) [31]	Operating

US data centers consumed 176 TWh of electricity in 2023, about 4.4% of total US electricity consumption. [42] The Lawrence Berkeley National Laboratory projects that figure could reach 6.7-12% by 2028, driven largely by AI workloads. [42]

Autonomous Evolution of EDA Tools: Multi-Agent Self-Evolved ABC

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Abstract

This paper introduces the first *self-evolving* logic synthesis framework, which leverages Large Language Model (LLM) agents to autonomously improve the source code of ABC, the widely adopted logic synthesis system. Our framework operates on the *entire integrated ABC codebase*, and the output repository preserves its single-binary execution model and command interface. In the initial evolution cycle, we bootstrap the system using existing prior open-source synthesis components, covering flow tuning, logic minimization, and technology mapping, but without manually injecting new heuristics. On top of this foundation, a team of LLM-based agents iteratively rewrites and evolves specific sub-components of ABC following our “programming guidance” prompts under a unified correctness and QoR-driven evaluation loop. Each evolution cycle proposes code modifications, compiles the integrated binary, validates correctness, and evaluates quality-of-results (QoR) on *multi-suite benchmarks including ISCAS 85/89/99, VTR, EPFL, and IWLS 2005*. Through continuous feedback, the system discovers optimizations beyond human-designed heuristics, effectively *learning new synthesis strategies* that enhance QoR. We detail the architecture of this self-improving system, its integration with ABC, and results demonstrating that the framework can autonomously and progressively improve EDA tool at full million-line scale.

Within this broader landscape, logic synthesis is a particularly representative domain. The *ABC* framework has emerged as the de facto academic and industrial research platform for synthesis and verification [14]. Over the past two decades, numerous research contributions, including cut-based mapping [5, 17], Boolean rewriting [15, 27], flow-tuning algorithms [19, 29], etc., have been proposed as extensions to *ABC*. However, most of these advances exist as external prototypes, loosely integrated scripts, or ad hoc binaries. Their capabilities often remain inaccessible within the monolithic *ABC* codebase itself, due to the complexity of integrating new heuristics, the rigidity of existing subsystems, and the high development barrier of its low-level C implementation. Even within the built-in algorithms, many heuristics (e.g., cut selection, refactoring conditions, choice-node expansion, and cost estimation) are statically designed by experts and rarely revisited, despite the underlying NP-hard nature of their decisions [30]. The result is a highly capable but difficult-to-evolve software system in which substantial QoR improvements may remain untapped.

Recent advances in code-generating AI, particularly Large Language Models (LLMs), provide a promising new perspective for automating EDA tool development [6, 25, 28]. OpenAI’s Codex and DeepMind’s AlphaCode have demonstrated that LLMs can generate functioning code and even solve complex programming challenges at a competitive level [3, 7]. Google DeepMind’s *AlphaE-*

“This paper introduces the first self-evolving logic synthesis framework, which leverages Large Language Model (LLM) agents to **autonomously improve the source code of ABC**, the widely adopted logic synthesis system. [...] Through continuous feedback, the system discovers optimizations **beyond human-designed heuristics**, effectively learning new synthesis strategies”

82v1 [cs.AR] 16 Apr 2026

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ARTICLE

The End of EDA: Why AI Will Eat the \$100B Chip Design Stack

by Palo Alto Electron • April 2026

For decades, hardware engineering has been mediated by tools. EDA stacks, PDKs, simulation environments — these defined not just workflows, but *who could participate* in the design process. Mastery meant fluency in these ecosystems. But that era is ending. Not because hardware is becoming simpler, but because **the interface to hardware is changing**.

A modern chip designer doesn't "draw transistors" in the traditional sense. They write constraints, configure flows, script simulations, interpret outputs, and iterate through abstractions. That is software thinking. The difference was that the environment was constrained — locked inside proprietary stacks, with high switching costs and institutional inertia.

AI breaks that constraint.

https://chiplet.us/news/the_end_of_eda_why_ai_will_eat_the_100b_chip_design_stack

2027: A Clear Fork in the Road

By the end of 2027, we will likely see two distinct classes of hardware engineers:

1. Tool Operators

- Depend on vendor flows
- Optimize within predefined boundaries
- Compete with automation

2. System Builders

- Create custom design infrastructure
- Orchestrate AI agents
- Integrate across domains (silicon → package → system)
- Define new capabilities, not just optimize existing ones

Only one of these groups compounds in value.

https://chiplet.us/news/the_end_of_eda_why_ai_will_eat_the_100b_chip_design_stack

What To Do Now

If you're in hardware today:

- Learn software beyond scripting — build systems
- Treat AI as a design collaborator, not a feature
- Start creating your own flows, even if imperfect
- Re-think IP as something you generate, not just license
- Move up the stack: from block → chip → system

https://chiplet.us/news/the_end_of_eda_why_ai_will_eat_the_100b_chip_design_stack

Final Thought

This isn't about replacing engineers. It's about redefining our craft. The next generation won't ask:

"Which EDA tool should I use?"

They'll ask:

"What system do I want to build — and how do I assemble the intelligence to build it?"

Hardware is not dying. It's becoming **programmable, composable, and intelligent**. And the engineers who embrace that shift won't just stay relevant —they'll define the next era of computing.

https://chiplet.us/news/the_end_of_eda_why_ai_will_eat_the_100b_chip_design_stack



This week

Monday

- Mapping deep neural nets (DNN/CNN) on GPUs
- Systolic arrays
- TPUs
- Transformers on GPUs

Wednesday

- Codefest #5
 - Systolic array
 - Milestone 2

SETH'S BLOG

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The steps vs. the concept

If you memorize the steps, you have a direct, simple and fast path to obtain the result.

Until the world changes.

Even the tiniest shift in the system will render your memorization useless.

On the other hand, if you understand the concept, you'll be able to produce the steps whenever you need them.

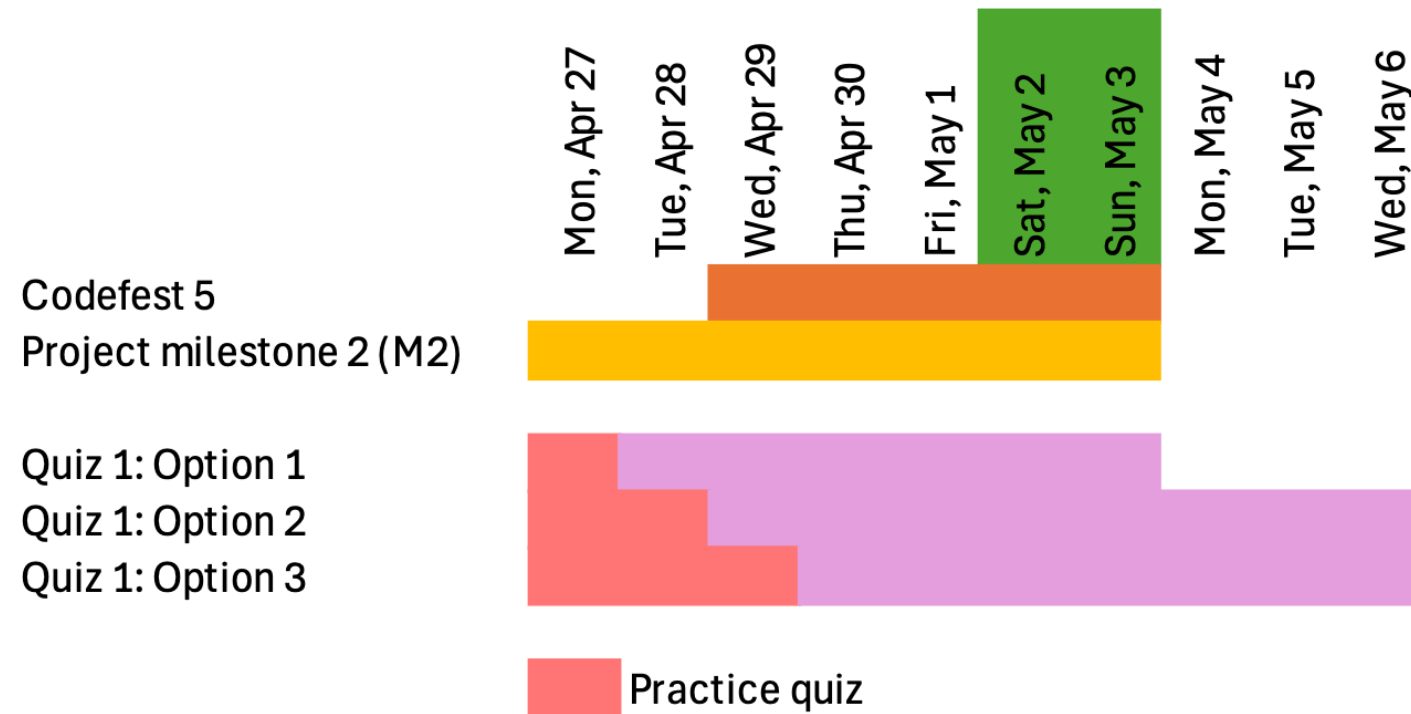
Quiz 1 format

- **Format:**
 - Online oral examination on the *Shine* platform.
 - I want to see your thinking and understanding of the foundational concepts. Do you know what you are talking about?
- **Duration:** 20min
- **Number of questions:** 5 questions, 4 min per question
- **Grading:** S/NY (80% threshold). A quiz 1 repeat will consume 2 tokens
- **Do the quiz 1 practice session now!**

<https://teuscher-lab.ece.pdx.edu/shine>



Quiz 1 scheduling



Look out for an e-mail with the exam PIN!